# Geometry side of the joint why-$\alpha$ magnitude integral (Grace, 2026-06-29). Computed Lyra's L1 Shilov-boundary step integrand I_k=$\int\{S^4\}\bar{Y}$\{k+1\}(u)·u_j·Y_k(u)d$\sigma$ explicitly ($S^4$ zonal = Gegenbauer C_k^\{3/2\}, weight $(1-t^2)$). RESULT: the per-step NORMALIZED amplitude I_k → EXACTLY 1/2 as k grows (0.478,0.488,0.492,…,→0.5) — the holo/antiholo HALF = the geometric meaning of the structural '2' in 2×6. The 6-step ladder (electron k=1 → first bulk discrete series k=C_2+1=7) gives ∏I_k≈1.4e-2, $\prod|I\_k|^2$≈2.0e-4 (plain $L^2$); Hardy-weight p=5/2 reweighting gives ~5e-8 / 2e-15. EVERY version is an O(geometric) number — NEVER $\alpha$^\{12\}≈2.3e-26 (off by 11+ orders). CONCLUSION (honest, robust to normalization): geometry fixes the 2×6 STRUCTURE (per-step→1/2 = holo/antiholo, ×6 steps = C_2) but CANNOT produce $\alpha$ — the magnitude $\alpha$-per-step must enter from the COUPLING (Elie's SO(4,2)/Sakharov EM side), NOT from any boundary overlap. The no-fake line is now GEOMETRICALLY CONFIRMED by explicit computation: the why-$\alpha$ is genuinely physics (coupling), not fakeable from $D\_IV^5$ geometry. Validates Lyra L3 (propagator forces 2×6) on the geometry side; delimits the open piece precisely for Elie E3 cross-check. No count move (9/26).

Grace

2026-06-29 Monday

## The Shilov-boundary ladder: geometry fixes 2×6 STRUCTURE, cannot carry $\alpha$

Joint why-$\alpha$ magnitude integral, geometry side (Lyra propagator + Elie heat-kernel + my geometry). Lyra posted the L1 integrand; I computed it explicitly. This is the no-solo-race piece done collaboratively now that the level-structure is on disk.

## What I computed

Lyra's L1 step integrand on the Shilov boundary $\check{S}=(S^4\times S^1)/Z_2$:

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I_k = ∫_{S⁴} Ȳ_{k+1}(u) · u_j · Y_k(u) dσ(u),  Y_k = degree−k S⁴ zonal harmonic = Gegenbauer C_k^{3/2}
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$S^4$ zonal harmonics are Gegenbauer C_k^{$\lambda$}, $\lambda$=(n−1)/2=3/2 (n=4), orthogonal w.r.t. $(1-t^2)$^{$\lambda$−1/2}=$(1-t^2)$ on [−1,1]. I computed the normalized step amplitude I_k = ⟨C_{k+1}|t|C_k⟩ / (||C_{k+1}|| ||C_k||) numerically.

## Result 1 — the per-step amplitude → EXACTLY 1/2 (the geometric "2")

| k | 1 | 2 | 3 | 6 | 10 | 20 | 40 |
|---|---|---|---|---|---|---|---|
| I_k | 0.4781 | 0.4880 | 0.4924 | 0.4971 | 0.4987 | 0.4996 | 0.4999 |

I_k → 1/2. A unit-coordinate multiplication splits evenly between raising (k→k+1) and lowering (k→k−1) as k grows → the up-leg carries 1/2. This is a clean geometric MAGNITUDE prefactor (≈1/2 per step).

PRECISION (self-correction): the per-step 1/2 is NOT the "2" in the exponent 2×6 — they are different objects. The exponent-2 is the sesquilinear/Born doubling (the mass uses $|\text{amplitude}|^2$, Lyra L2(b)); the 1/2 is a raise/lower magnitude prefactor. So: exponent = 2 (sesquilinear) × 6 (C_2 steps); magnitude prefactor = (≈1/2)^6 per leg (geometric, O(1)). Don't conflate the magnitude-1/2 with the exponent-2.

## Result 2 — the 6-step ladder magnitude is O(geometric), NEVER $\alpha$

Electron k=1 → first bulk discrete series k=C_2+1=7 is C_2=6 steps (Lyra L2 target-innocent count):

| quantity | value |
|---|---|
| $\prod$ I_k (k=1..6), amplitude (plain $L^2$) | $1.4\times10^{-2}$ |
| $\prod$ $|I\_k|^2$ (sesquilinear) | $2.0\times10^{-4}$ |
| Hardy-weight p=5/2 reweighted $\prod|I\_k|^2$ | $2.2\times10^{-15}$ |
| $\alpha^{12}$ (target) | $2.3\times10^{-26}$ |

Every normalization gives an O(geometric) number — none is $\alpha^{12}$. The plain-$L^2$ product is $\approx(1/2)^6$-ish; the Hardy reweight pushes it down but still 11 orders off $\alpha^{12}$. The conclusion is robust to the normalization convention (which is Lyra's propagator territory): no geometric measure on $D\_IV^5$ contains $\alpha$.

## The honest conclusion (no-fake line, geometrically confirmed)

Geometry fixes the 2×6 STRUCTURE — per-step → 1/2 (holo/antiholo, the "2"), six steps (k=1→7, C_2 = the "6") — but CANNOT produce the magnitude $\alpha$. $\alpha$ is a coupling; it lives in the EM charge attached to the $S^1$ phase per step (my G2: e^{i$\varphi$} = the charge quantum per level), i.e. in Elie's SO(4,2)/Sakharov coupling side, not in the $S^4$ overlap. This is the no-fake line I flagged Monday (working_state line 15) — now confirmed by explicit computation: the why-$\alpha$ magnitude is genuinely physics (the per-step coupling), not fakeable from $D\_IV^5$ geometry.

This is the right kind of negative: it does not block the derivation, it localizes it. The open piece is exactly "what is the EM coupling amplitude per ladder step?" (Elie E2/E3, Lyra propagator), and the geometry guarantees that whatever it is, the count is 2×6 and the geometric prefactor is O(1).

## Addendum — the exact per-step values, and why they REINFORCE the no-fake line ((C), disciplined)

Exact closed form (derived): $I\_k^2 = (1/4)\cdot(k+1)(k+3)/((k+3/2)(k+5/2))$. The first values:

| k | $I\_k^2$ | clean? |
|---|---|---|
| 1 (electron) | $8/35 = \text{rank}^3/(n\_C\cdot g)$ | clean (target-innocent geometric number) |
| 2 | $5/21 = n\_C/(N\_c\cdot g)$ | clean |
| 3 | $8/33 = 8/(3\cdot 11)$ | 11 not a substrate primary |
| 4 | $35/143 = 35/(11\cdot 13)$ | not clean |
| 5 | $16/65 = 16/(5\cdot 13)$ | not clean |
| 6 | $21/85 = 21/(5\cdot 17)$ | not clean |

Honest tiering ((C), NOT a win): the two LOWEST steps land on substrate-clean fractions (small integers 2,3,5,7), but the cleanness does not persist — k≥3 hit non-primary 11, 13, 17. So $I\_1^2=\text{rank}^3/(n\_C\cdot g)$ is a clean low-k geometric identity but not a ladder-wide structure; it's the small-integer coincidence you expect from generic Gegenbauer ratios. This REINFORCES the no-fake line: even the geometric prefactors are generic O(1) Gegenbauer numbers, only coincidentally substrate-clean at the lowest steps — there is no deep substrate magnitude hiding in the overlap, confirming $\alpha$ must come entirely from the coupling. (Cal #286 lens: rich-vocab, partial mechanism, non-persistent → (C). Recorded, not banked.)

— Grace, 2026-06-29 Monday. Geometry side of joint why-$\alpha$: I_k→1/2 (the "2"), 6 steps (C_2), magnitude O(geometric) NEVER $\alpha$ → $\alpha$-per-step is COUPLING-physics (Elie), no-fake line geometrically confirmed. Lyra L3 validated on geometry side. For Elie E3 cross-check. No count move (9/26).